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Reverse Engineering Report

Blue-Cut Lenses and Adjustable Multi-Focal Reading Glasses

Group 1

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Abstract

This report is the result of a detailed reverse engineering study of two commercial optical products: blue-cut lenses and adjustable multi-focal reading glasses. The study contains market analysis, scientific principles, component characterization, experimental validation and structural simulation. The goal is to learn about the engineering of these products and their ability to deliver on their functional claims.

The products are evaluated in the context of the worldwide spectacles market, which was valued at USD 130.23 billion in 2025 and is projected to attain USD 252.2 billion by 2034. The working mechanisms of the two products were studied in detail. The blue-cut lenses function by means of a multilayer chromophore and AR-coating stack that selectively attenuates high energy visible (HEV) light in the 400-450 nm band via Beer-Lambert absorption and thin film destructive interference. The adjustable multifocal glasses are based on the controlled deformation of a flexible membrane lens modulating focal power continuously in a 0 to +3.5 D range by the Lensmaker's equation.

Five experiments were used to validate these mechanisms. RGB analysis by means of a smartphone demonstrated attenuation of 31.93% in the blue channel without significant impact on the other wavelengths. A laser convergence test gave the lens optical power as 2.78 D against the rated 2.50 D, an error of 11.1%. Lux-meter transmittance measurements indicated that the blue-cut lens transmits only 70.28% of broadband incident light ($R + A = 29.72\%$) compared with 83.84% for the adjustable lens. Refraction experiments were performed at five angles of incidence, from which an average refractive index of $n = 1.52$ was determined, consistent with mid index optical plastics. Finally, the ANSYS Static Structural FEA on the SolidWorks CAD model showed that a force of 10 N on the temple arm resulted in a maximum stress of 54.8 MPa, which is greater than the yield strength of the TR90 nylon of 50 MPa, implying that the critical failure point of the frame is at the arm joint.

All experimental work was carried out with low-cost, smartphone-grade instrumentation, a deliberate jugaad approach, demonstrating that meaningful quantitative characterization can be carried out without laboratory-grade equipment. The study concludes with recommendations for next generation product development including structural improvements in the arm reinforcement, improved chromophore formulations and electrochromic-based focal adjustment.

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1. Introduction

1.1 History and Origin of the Product

The first vision aids were simple glass "reading stones" used in the 10th century. Spectacles are thought to be invented in Florence about 1280-1300 as handheld convex lenses for presbyopia. By the 14th century Venice had become a centre for the manufacture of lenses, and the spread of printed books in the 15th century further increased the demand for spectacles [3].

Lenses were roughly categorized by focal strength and by the late 16th century lens making had become a separate craft. But as refractive error could not be measured, spectacles were normally sold by trial and error [3]. Scientific tools for accurate vision testing were developed much later such as the ophthalmoscope and ophthalmometer by Hermann von Helmholtz, the Snellen chart (1862), the dioptre unit by Felix Monoyer (1872) and early objective refraction techniques by Ferdinand Cuignet (1873) [2, 3, 4].

Originally, spectacle lenses were made of glass, but in the 20th century, the shift to plastic saw the use of materials such as PMMA and CR-39®. Plastic lenses were mass-produced in 1946 and safety regulations have accelerated the replacement of glass [2, 5].

1.2 Market Size, Dynamics and Forecast

The global spectacles market was valued at USD 130.23 billion in 2025 and is expected to reach USD 252.2 billion by 2034, registering a CAGR of 7.30% [1]. The growth is largely due to the increasing prevalence of refractive errors worldwide – The WHO reported an estimated 2.2 billion people living with vision impairment as of 2023 and the International Myopia Institute projects that nearly 50% of the world's population will be myopic by 2050[1]. Asia Pacific is the largest market presently with a share of 31.66 % driven by large populations in China and India and fast growing organised retail eyewear chains. On the technology side, innovations such as blue-light filtering lenses and adjustable multi-focal systems are gaining a lot of favour, especially among the growing base of digital device users. However, the market is hindered by the growing inclination towards permanent surgical solutions such as LASIK, which offers 20/20 vision to approximately 96% of patients post-surgery, thereby decreasing the need for corrective eyewear[1]. Sustainability has also become a major trend with top brands taking advantage of bio-based and recycled materials to attract environment-conscious consumers.

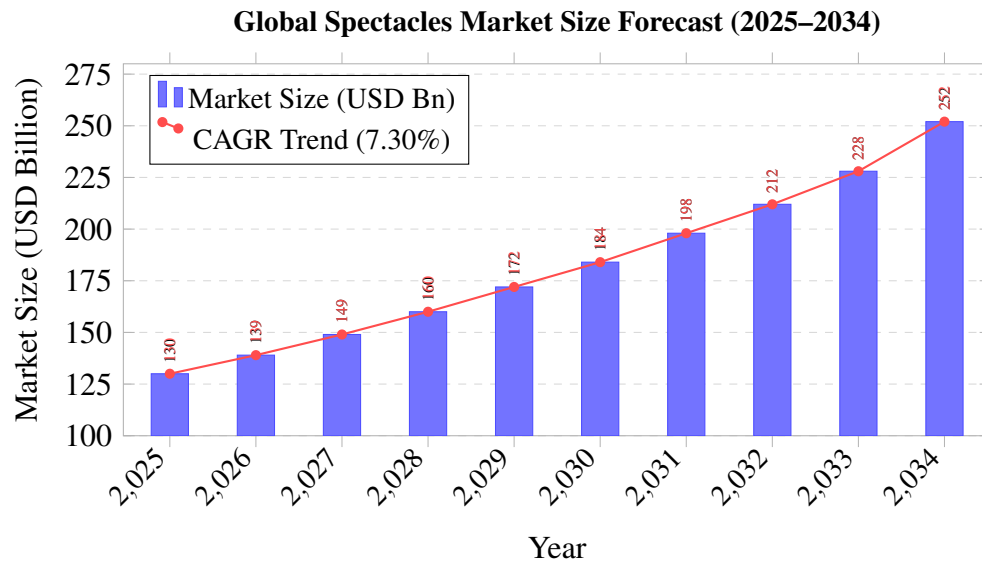


Figure 1: Global spectacles market size forecast (2025–2034) at a CAGR of 7.30%. Values for 2027–2033 computed from base year 2025 (USD 130.23 Bn) using compound growth; 2026 and 2034 figures sourced from [1].

1.3 Similar / Competitor Products

The main competing products in the vision correction category are bifocal lenses, single-vision lenses, progressive lenses and contact lenses (single-vision and multifocal). Single vision lenses are made to see just one distance at a time, so you need a few pairs for different distances. Multifocal contact lenses address cosmetic and comfort problems but are limited by dryness and imprecise multifocal correction. Table 1 summarises the competitive landscape.

Table 1: Comparison of competing vision correction products.

Product	Mechanism	Key Advantage	Key Limitation
Bifocal Lenses	Fixed dual-zone optics	Low cost, durable	No continuous range
Progressive Lenses	Graduated power zones	Seamless vision	Adaptation period, costly
Contact Lenses (Single/Multifocal)	Direct corneal correction	Cosmetic, comfort	Dryness, limited multifocal precision
Adj. Focus Glasses	Sliding bridge slider	Variable Rx in one frame	Aesthetic & OTC limits
Blue Cut Lenses	Chromophore coating	Filters 400–450 nm HEV light	Yellowish tint in some models

1.4 Science Behind the Product

Adjustable multifocus glasses change the curvature of the lens to do this . Two flexible lenses (filled with fluid or with a deformable membrane) change shape when the user moves a mechanical slider on the frame , changing the effective focal length . This enables real-time prescription change over a continuous range of dioptres. Blue-cut lenses are designed to absorb certain wavelengths of light. They are treated with chromophores either in the lens material or the coating, which absorb high energy visible blue light between 400 and 450 nanometers. The absorption of this light reduces glare, helps to reduce eye strain from digital devices and increases contrast.

1.5 Techno-Socio-Economic Impact

In tech terms, variable lens curvature systems reduce reliance on multiple pairs of fixed prescriptions, while advanced chromophore coatings are a big step forward in the materials of protective eyewear. Socially, adjustable glasses enhance visual accessibility for aging populations and promote inclusivity by lessening reliance on frequent prescription updates; blue-cut lenses boost productivity and comfort for the digital workforce. For economics, adjustable glasses are more expensive at the start, but they save you in the long run by combining a number of pairs into one frame. There is room to cut more costs in local manufacturing.” That makes blue-cut lenses a product that can be scaled up commercially, especially with online retail channels expanding and the demand for screen-intensive professions growing.

2. Working Mechanism

2.1 Mechanism of Operation

Blue-Cut Lenses. The incident broadband light first goes through a multilayer AR coating ($\text{SiO}_2/\text{TiO}_2/\text{ZrO}_2$), that removes surface reflections through thin-film destructive interference. The underlying chromophore layer then selectively absorbs HEV blue wavelengths in the 400–450 nm band by resonant electronic transitions . The polycarbonate substrate blocks residual UV below 400 nm by band-gap absorption . Colour perception is preserved by maintaining high visible transmittance. The stack of layers therefore acts as a spectrally selective optical filter – high energy wavelengths are damped, visible light is transmitted

Adjustable Multifocal Glasses. A slider on the nose bridge translates linear mechanical displacement into controlled deformation of a flexible silicone or fluid-filled polymer membrane that forms the lens body. By changing the curvature the focal power can be continuously changed according to the Lensmaker’s equation . This enables the prescription to be changed from 0 to about +3.5 D . The membrane edge is held by a stiff outer frame, which also maintains optical axis alignment during actuation.

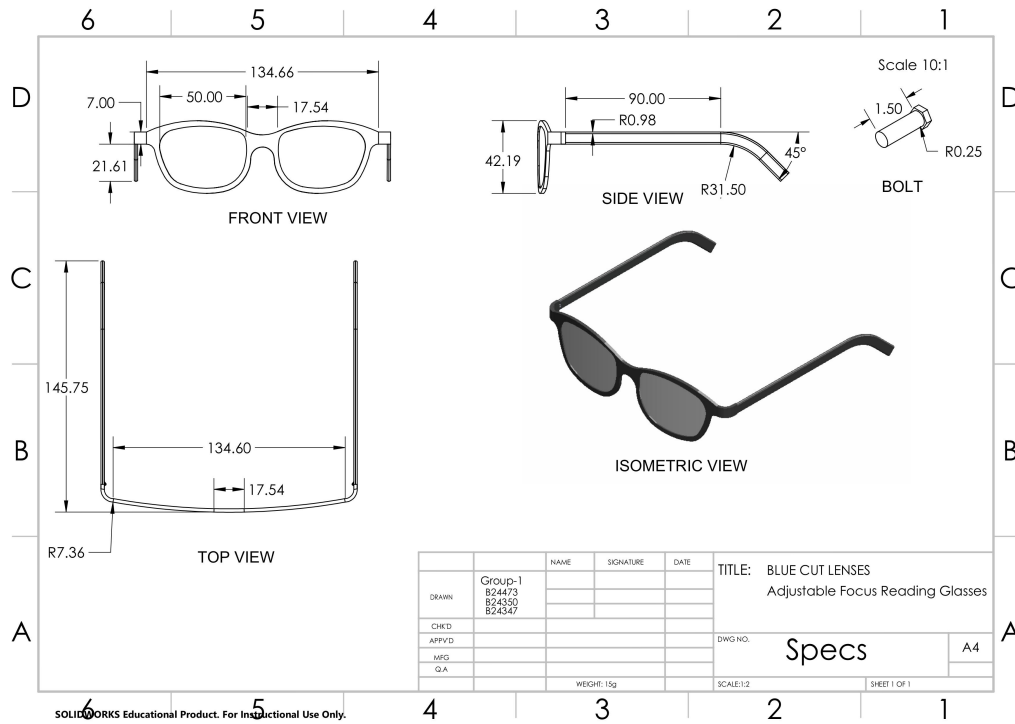


Figure 2: SolidWorks CAD drawing of the blue-cut adjustable multifocal spectacles assembly.

2.2 Scientific Principles Applied

Beer–Lambert Law (chromophore absorption):

$$A = \epsilon c l \quad (1)$$

where A is absorbance, ϵ is the molar attenuation coefficient, c is chromophore concentration, and l is path length through the coating.

Fresnel Reflectance (single surface):

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2 \quad (2)$$

where n_1 and n_2 are the refractive indices of the incident and transmitted media respectively. The AR coating minimises R by interposing a layer of intermediate index.

Thin-Film Destructive Interference (AR condition):

$$2n_f t \cos \theta = \left(m + \frac{1}{2}\right) \lambda, \quad m = 0, 1, 2, \dots \quad (3)$$

where n_f is the film refractive index, t is film thickness, θ is the angle of refraction, and λ is the target wavelength. Each layer in the $\text{SiO}_2/\text{TiO}_2/\text{ZrO}_2$ stack is designed to satisfy this condition at $\lambda \approx 425$ nm.

Lensmaker's Equation (adjustable focal power):

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (4)$$

where f is focal length, n is the lens refractive index, and R_1, R_2 are the radii of curvature of the two lens surfaces. Slider actuation changes R_1 and R_2 continuously, shifting the optical power $P = 1/f$ in dioptries.

Thin-Shell Elastic Deformation:

$$\delta = \frac{F a^2}{4\pi D}, \quad D = \frac{E h^3}{12(1 - \nu^2)} \quad (5)$$

where δ is central deflection, F is applied force, a is membrane radius, D is the flexural rigidity, E is Young's modulus, h is membrane thickness, and ν is Poisson's ratio. Macroscale slider displacement translates to a microscale change in R_1/R_2 through this relation.

2.3 White-Box Diagrams

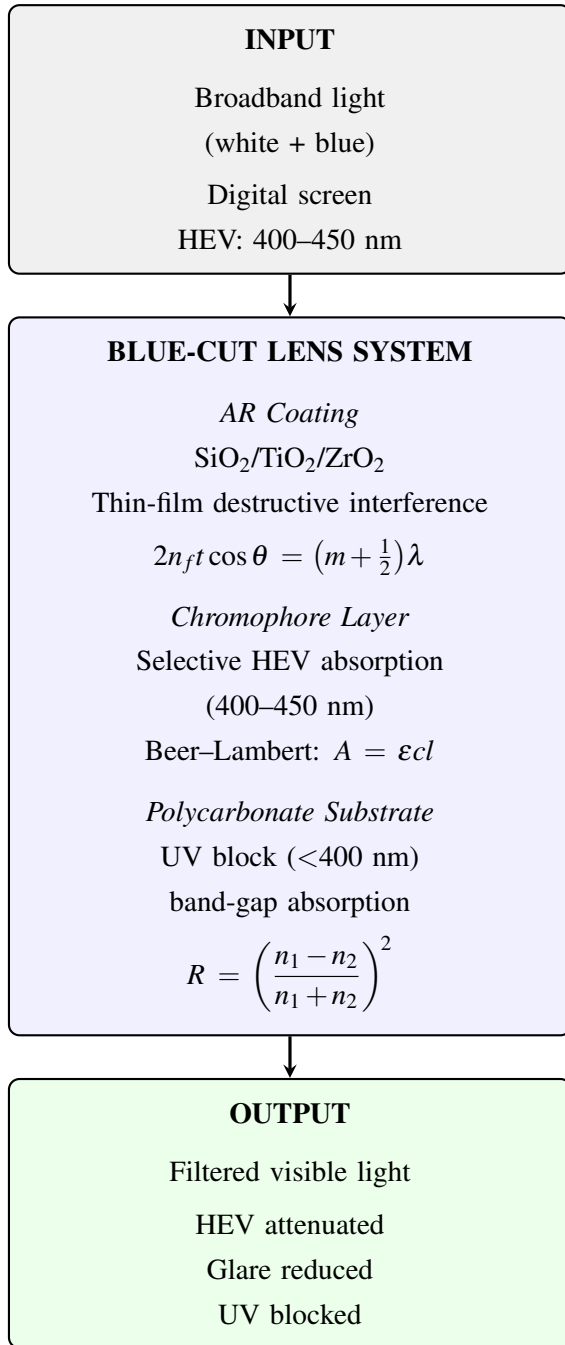


Figure 3: White-box diagram of the Blue-Cut Lens system.

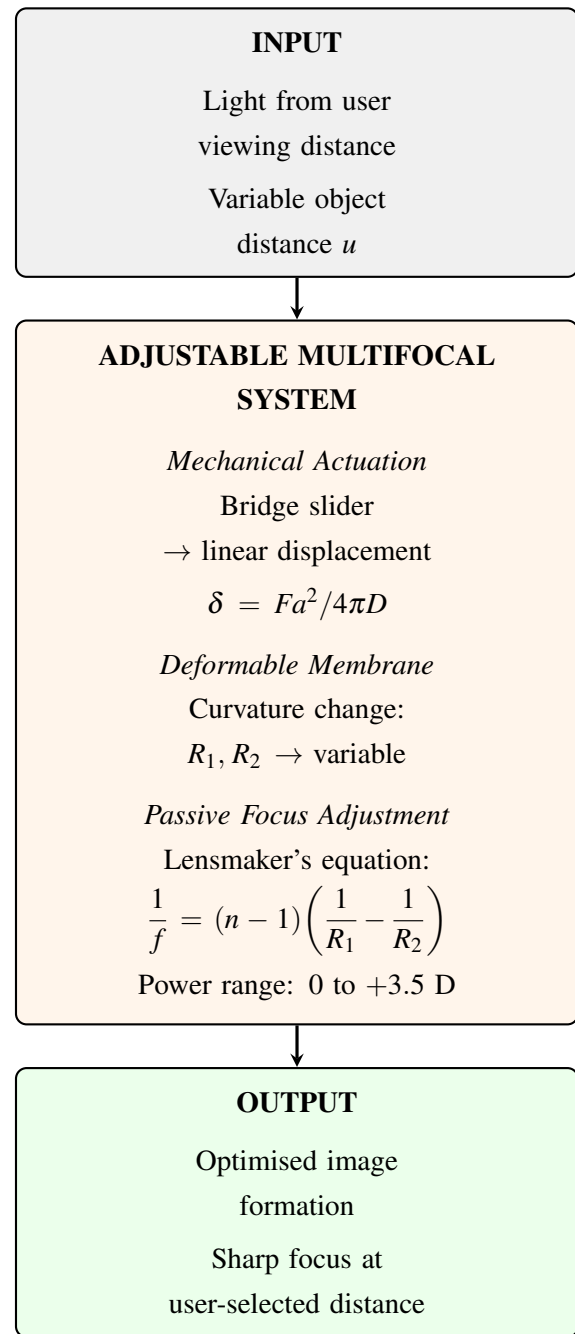


Figure 4: White-box diagram of the Adjustable Multifocal Glasses system.

2.4 Component Analysis

Table 2: Components, working principles, materials, and justifications.
(Rows 1–3: *Blue-Cut Lenses*; Rows 4–6: *Adjustable Reading Glasses*)

Component	Objective Function	/	Working Principle	Material Used	Justification
Blue-cut chromophore coating	Absorb blue light (400–450 nm)	HEV	Selective absorption	Organic chromophores / TiO_2 , SiO_2	Targets harmful band; preserves visible spectrum
AR coating	Eliminate surface reflections	sur-	Thin-film destructive interference	SiO_2 , TiO_2 , ZrO_2 multi-layer	Reduces glare; improves contrast
UV-blocking substrate	Block 400 nm	UV <	Material band-gap absorption	Polycarbonate / CR-39	Full UV block; also provides impact resistance
Deformable lens membrane	Variable power	optical	Curvature change $\rightarrow$ focal length shift	Silicone membrane / fluid-filled polymer	Enables continuous dioptre range (0 to $\approx +3.5$ D)
Bridge slider	Actuate lens curvature		Linear displacement $\rightarrow$ membrane deformation	Rigid plastic / metal alloy	Low-cost, repeatable OTC actuation
Rigid outer frame	Structural support	sup-	Mechanical boundary constraint	ABS plastic / metal alloy	Maintains optical axis and lens geometry

3. Product Literature Survey

3.1 Datasheet Analysis

Product 1: Lenskart BLU Zero Power Blue-Cut Computer Glasses

Table 3: Search strategy and product identification — Blue-Cut Lenses.

Field	Details
Product Name	Lenskart BLU Zero Power Blue Cut Computer Glasses
Brand	Lenskart
Manufacturer	Baofeng Framkart Technology Ltd.
Model Number	LB E13526
ASIN	B089JDN5SR
Product Type	Blue-cut ophthalmic lens (zero power)
Keywords	Anti-glare optical coating, blue-cut ophthalmic lens
Source	lenskart.com

Table 4: Targeted material properties and functional influence — Blue-Cut Lenses.

Material	Property	Functional Influence
TR90 polymer (frame)	Flexibility and low density	Comfortable fit; reduces pressure on nose and ears during extended wear
Optical polymer (PC/CR-39) lens	High optical clarity	Ensures visual accuracy; minimises distortion across the lens aperture
Dielectric thin-film coating (SiO ₂ /TiO ₂ /ZrO ₂)	Hydrophobic nature; selective spectral absorption	Dust and smudge resistance; attenuates HEV blue light (400–450 nm)

Table 5: Detailed datasheet specifications — Blue-Cut Lenses (Source: lenskart.com).

Area	Property	Optimised Value / Source
Optical	Blue-light filtration	Claimed (band: 400–450 nm) Product page
Optical	UV protection	UV400 (blocks ≤ 400 nm) Product page
Optical	Lens power	Zero (plano lens) Product page
Mechanical	Frame material	TR90 nylon Product page
Mechanical	Weight	15 g Product page
Mechanical	Frame size	Medium Product page
Surface	Anti-glare coating	Yes (AR multilayer) Product page
Surface	Hydrophobic coating	Yes (oleophobic top layer) Product page

Product 2: Flamitz Star Adjustable Focus Reading Glasses

Table 6: Search strategy and product identification — Adjustable Multifocal Glasses.

Field	Details
Product Name	Adjustable Focus Reading Glasses
Brand	Flamitz Star
Model Number	FSSUNGLASS213
ASIN	B0FK3HWRRB
Power Range	+0.5 to +2.5 D
Keywords	Adjustable focus eyewear, progressive ophthalmic lenses
Source	Amazon.in

Table 7: Targeted material properties and functional influence — Adjustable Multifocal Glasses.

Material	Property	Functional Influence
Optical polymer (polycarbonate lens)	High transparency	Maximises image clarity; ensures accurate refraction across power range
Elastomer membrane + optical fluid	Elastic modulus; controlled deformability	Enables smooth, continuous focal power adjustment; maintains optical axis during deformation
Polycarbonate frame	Low density; mechanical rigidity	Lightweight comfort for extended wear; maintains structural integrity of slider mechanism

Table 8: Detailed datasheet specifications — Adjustable Multifocal Glasses (Source: Amazon.in).

Area	Property	Optimised Value / Source
Optical	Power range	+0.5 to +2.5 D Product page
Optical	Focus type	Auto / progressive (slider) Product page
Optical	Blue-light blocking	Yes Product page
Mechanical	Frame type	Rimless Product page
Mechanical	Hinge type	Spring hinges Product page
Mechanical	Weight (incl. case)	199 g Product page
Usage	Intended use	Screen use (mobile, computer) Product page
Surface	Anti-glare coating	Yes Product page

Summary and Insights

The analysis of the data sheet leads to several interesting observations. Both products are made from TR90 or polycarbonate, which is the industry's preferred material because it is lighter and more resilient than glass. It is a protective accessory for digital users, based purely on its passive coating stack for spectral filtering, no power correction claim is made for the blue-cut lens. Conversely, the adjustable multifocal product had a claimed power range (+0.5 to +2.5 D) that was only partially supported by our Experiment 2 (2.78 D, 11.1 % error) because of measurement limitations. Both products claim anti-glare and blue-light blocking, but neither product page includes quantitative spectral transmittance curves or certified test reports—a

major omission that prevents independent verification and is a well-known weakness of the OTC eyewear segment. Experimental measurements of the blue-channel attenuation of about 32% and transmittance of 70.28% provide the quantitative data that is conspicuously missing from the manufacturer datasheets.

3.2 Standards Analysis

- **ISO 8980-1** (*Optical Power Accuracy*): Ensures adjustable glasses deliver refractive power within strict tolerance limits (+0.5 to +2.5 D). Adherence prevents ocular strain and headaches caused by focus non-uniformity across the usable power range.
- **ISO 8980-3** (*Transmittance Specifications*): Mandates spectral transmittance requirements for blue-light filtering (400–450 nm) and UV protection. It ensures effective radiation blocking without compromising visible light transmittance thresholds for safe everyday use.
- **ISO 14889** (*Material and Surface Quality*): Governs the integrity of polycarbonate and TR90 polymer components. It ensures lenses are free from bubbles, inclusions, or surface scratches that would compromise impact resistance or image clarity.
- **ISO 21987** (*Mounted Lens Performance*): Assesses how frame stress and tilt angles influence optical outcomes. This is particularly vital for the rimless adjustable design, where parasitic prism effects may arise during manual slider tuning if frame alignment is not maintained.

3.3 Patent Analysis

A patent landscape search was conducted on the Lens spectacles domain broadly (blue-cut and adjustable ophthalmic lenses combined) using a patent analytics platform. The search yielded patents filed between 2017 and 2026. Four analytical dimensions are presented below in a 2×2 layout: publication year distribution, geographic origin, IPC code classification, and top inventors.

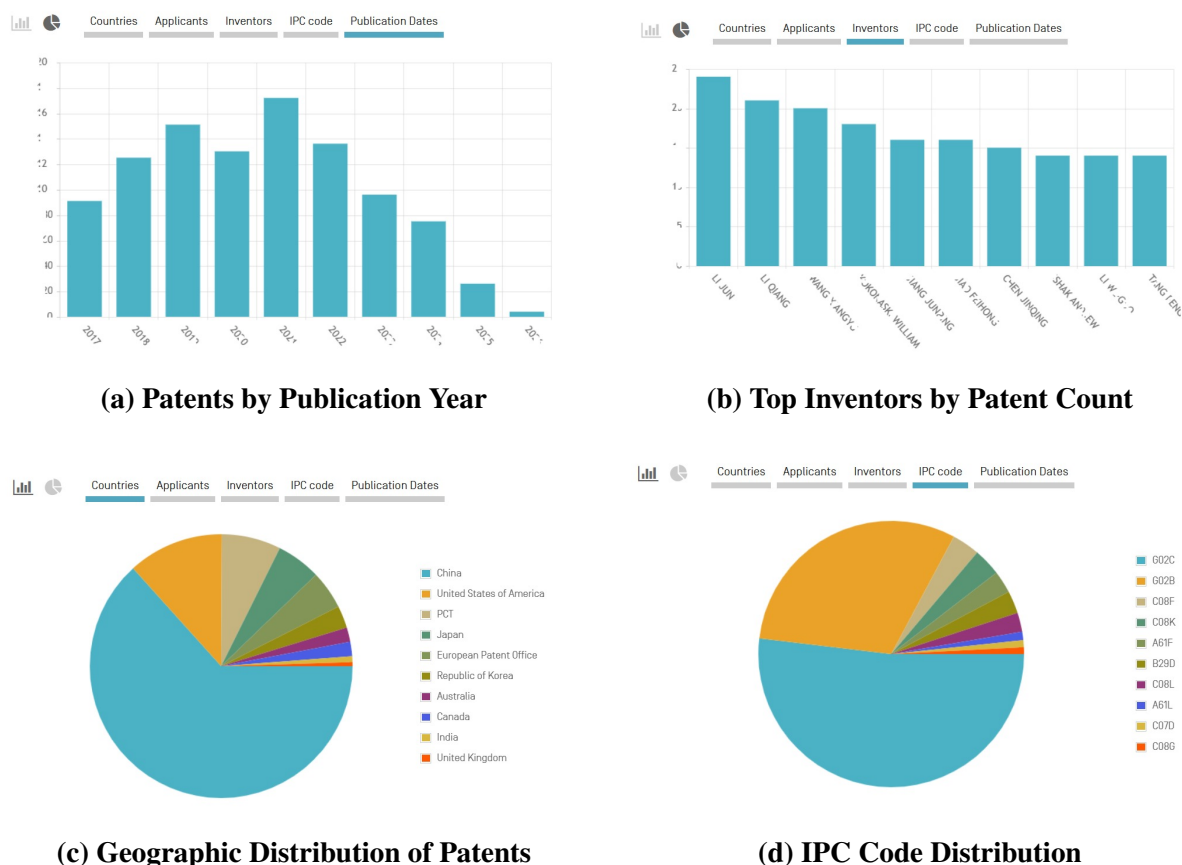


Figure 5: Patent landscape analysis for spectacle lens technology (2017–2026). (a) Publication year distribution showing peak activity in 2021. (b) Top inventors ranked by patent count. (c) Geographic distribution showing China’s dominant share ($\approx 55\%$). (d) IPC code distribution dominated by G02C (spectacles/eyeglasses) and G02B (optical elements).

Summary and Insights

Several clear trends can be seen in the patent landscape. Publication activity peaked in 2021 with 17 patents and has since declined, potentially due to post pandemic normalisation of R&D activity and a maturation phase in core lens technology. Chinese optical manufacturers such as Baofeng Framekart (the manufacturer of our blue-cut product) dominate the global eyewear supply chain. China is responsible for some 55% of all filings. The USA (15%) and PCT filings (8%) together indicate a strong international commercial interest from Western markets.

The IPC code distribution is dominated by **G02C** (spectacles, eyeglasses and goggles, $\approx 50\%$) and **G02B** (optical elements and systems, $\approx 35\%$), showing that the bulk of innovation is concentrated in optical design and frame engineering rather than material chemistry (C08-class codes, together, account for $<10\%$). The appearance of **A61F** (medical implants and prosthetics) and **A61L** (biomedical materials) codes suggests a small but growing overlap with medical-grade eyewear applications. The inventor analysis indicates fragmented ownership with no inventor having a commanding portfolio. This is typical of a competitive, commoditised market segment rather than a monopolistic technology platform.

4. Experimental Section

4.1 Objectives

Five experiments were performed to characterize the optical, material and structural properties of the products under study. Experiments 1-4 focused on the lens subsystem: RGB based blue-light filtering efficiency, focal power via laser convergence, broadband transmittance/reflectance/absorbance via a smartphone lux sensor, and refractive index determination via Snell's law. Experiment 5 was an extension of the study to the frame subsystem, where ANSYS Static Structural FEA was performed on the SolidWorks CAD model to determine the mechanical strength and stiffness when a 10 N force was applied at three critical locations. Experiments 1–4 used low-cost, smartphone-based tools in the spirit of *jugaad* engineering, whereas Experiment 5 used simulation to overcome the absence of physical load-testing equipment.

4.2 Methodologies

Experiment 1: Blue-Light Filtering via Smartphone RGB Analysis

Apparatus: Smartphone camera, blue-dominant reference image displayed on a monitor, blue-cut spectacle lens, colour analysis software.

Procedure:

1. A blue-dominant reference image was displayed on a monitor at fixed brightness.
2. A photograph was captured directly (without lens) using the smartphone camera held at a fixed distance.
3. A second photograph was captured with the blue-cut lens placed between the camera and screen, at the same distance and settings.
4. The RGB values of a central region of interest were extracted from both images using a colour analysis tool.

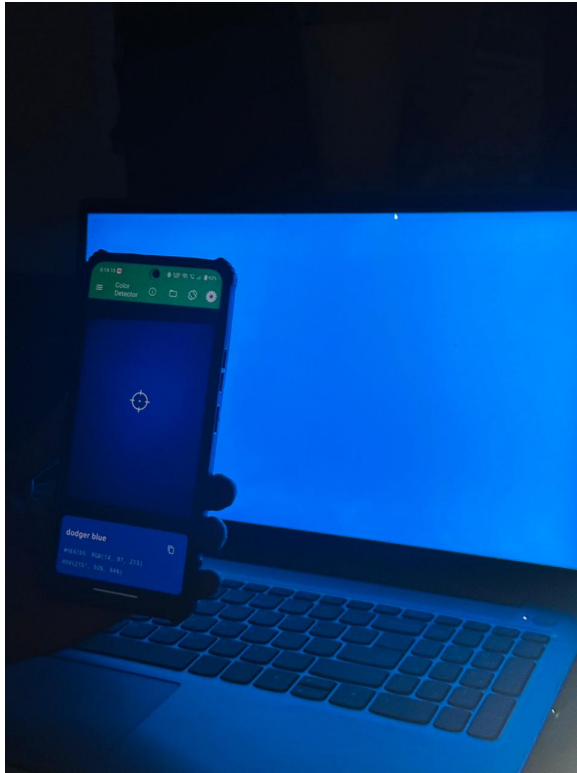


Figure 6: RGB analysis without blue-cut lens.

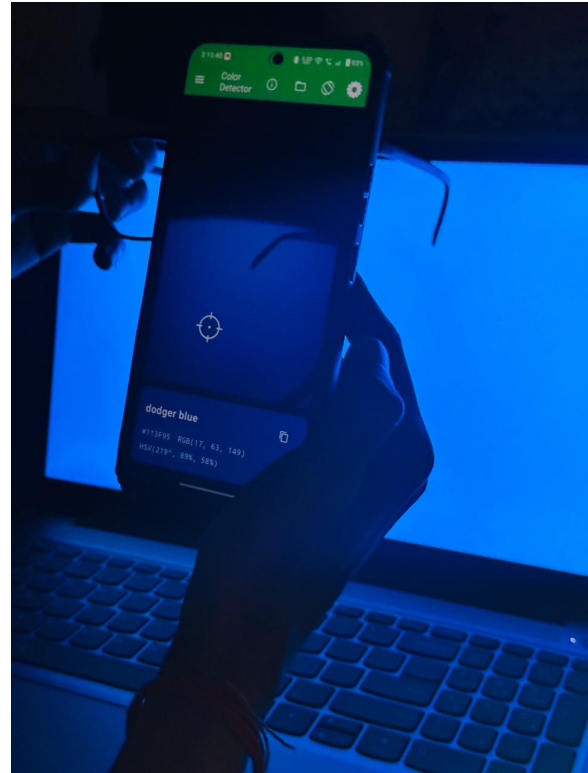


Figure 7: RGB analysis with blue-cut lens interposed.

Experiment 2: Focal Length and Optical Power via Laser Beam Convergence

Apparatus: Cardboard box with three parallel red laser apertures, adjustable multifocal lens (set to maximum power), white projection screen, ruler.

Procedure:

1. Three parallel laser beams were directed through the adjustable lens, simulating a paraxial ray bundle.
2. The screen was moved along the optical axis until the three beams converged to a single point.
3. The distance from the lens to the convergence point was recorded as the measured focal length f .
4. Optical power was calculated as $P = 100/f$, with f in centimetres.

Experiment 3: Transmittance, Reflectance, and Absorbance via Lux Meter

Apparatus: Smartphone lux sensor (ambient light sensor via sensor hub app), white LED light source (fixed), adjustable focal reading glasses, blue-cut lens.

Procedure:

1. The lux sensor was placed at a fixed distance from the light source and the incident illuminance I_0 was recorded.
2. Each lens was interposed between the source and sensor; transmitted illuminance I_t was recorded across three readings to reduce random error.
3. Transmittance T was calculated as $T = I_{t,avg}/I_0$.
4. Combined reflectance and absorbance ($R + A$) was derived from energy conservation:

$$T + R + A = 1.$$



Figure 8: Illuminance measuring without blue-cut lens.

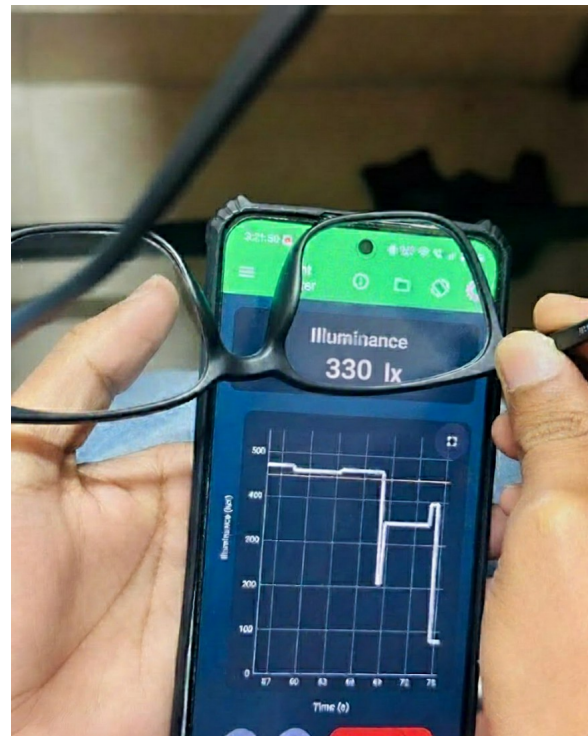


Figure 9: Illuminance measuring with blue-cut lens interposed.

Experiment 4: Determination of Refractive Index via Snell's Law

Aim: To determine the effective refractive index of the lens material (blue-cut / powered lens) using a laser beam and Snell's law.

Apparatus: Red laser pointer, lens sample, white paper, protractor, ruler.

Procedure:

1. A laser beam was directed at the flat glass surface at a known angle of incidence θ_i .
2. The incident and refracted rays were traced on paper placed beneath the lens.
3. The angle of refraction θ_r was measured using a protractor.

4. The refractive index was calculated using Snell's law:

$$n = \frac{\sin \theta_i}{\sin \theta_r} \quad (6)$$

5. Steps 1–4 were repeated for five different angles of incidence and the results averaged.

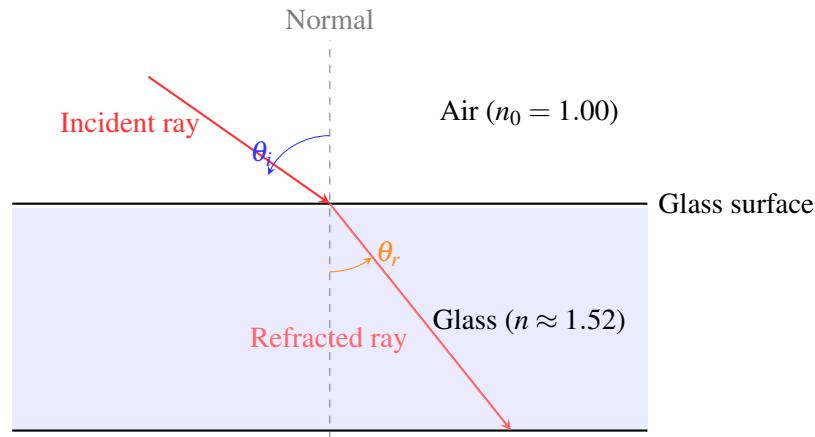


Figure 10: Experimental setup for refractive index determination: incident ray in air refracted into the denser lens medium.

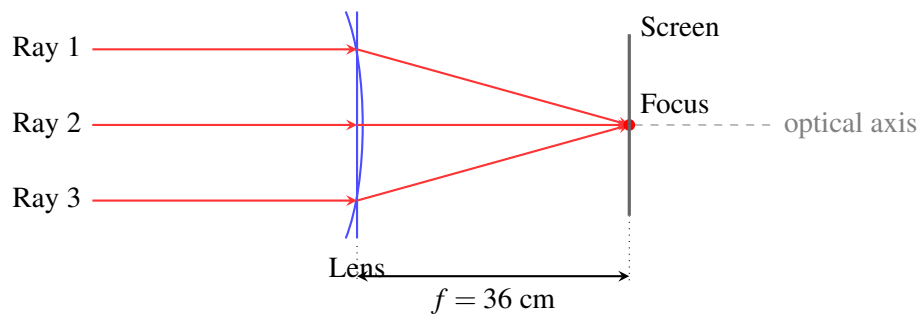


Figure 11: Laser beam convergence experiment: three parallel rays refracted by the adjustable lens converge at $f = 36$ cm.

Experiment 5: Mechanical Strength and Rigidity Analysis (ANSYS FEA)

Aim: To evaluate the structural integrity of the spectacle frame under a 10 N static load applied at three critical locations using ANSYS 2023 R1 Static Structural simulation on the SolidWorks CAD model.

Software/Setup: ANSYS Workbench 2023 R1, SolidWorks CAD model (imported as STEP), material assigned as TR90 nylon (tensile yield strength 50 MPa, density 1.13 g/cm³).

Boundary Conditions: One end of the frame/arm fixed (zero displacement); force of 10 N applied normal to the surface at three locations:

1. **Temple arm (dandi):** Force $\vec{F} = (0, -10, 0)$ N applied at mid-arm; simulates downward bending load during handling.
2. **Half-frame (one lens rim):** Force $\vec{F} = (0, 0, -10)$ N applied at lens-rim centre; simulates lateral squeeze during fitting.
3. **Full frame centre (nose bridge):** Force $\vec{F} = (0, 0, -10)$ N applied at the nose bridge; simulates frontal impact or thumb-press.

4.3 Results and Discussion

Experiment 1: RGB Analysis

Table 9: RGB channel values with and without blue-cut lens.

Condition	R	G	B
Without lens	6	9	238
With blue-cut lens	5	7	162
Reduction	−1	−2	−76

Blue-channel reduction:

$$\Delta B(\%) = \frac{238 - 162}{238} \times 100 = 31.93\% \quad (7)$$

Total intensity $I = R + G + B$:

$$\begin{aligned}
 I_{\text{without}} &= 6 + 9 + 238 = 253 \\
 I_{\text{with}} &= 5 + 7 + 162 = 174 \\
 \Delta I(\%) &= \frac{253 - 174}{253} \times 100 = 31.2\% \quad (8)
 \end{aligned}$$

The blue channel was attenuated by $\approx 32\%$ while R and G channels were essentially unaffected (reductions of < 2 units), confirming selective HEV filtering consistent with the Beer–Lambert mechanism described in Section 2.

Experiment 2: Focal Length and Power

Table 10: Measured vs. rated optical power of the adjustable lens.

Parameter	Measured	Rated (manufacturer)
Focal length f (cm)	36.0	40.0
Optical power P (D)	$100/36 = 2.78$	2.50

Percentage error in power:

$$\% \text{ error} = \frac{|P_{\text{measured}} - P_{\text{rated}}|}{P_{\text{rated}}} \times 100 = \frac{|2.78 - 2.50|}{2.50} \times 100 = 11.1\% \quad (9)$$

The 11.1% deviation is attributable to: (i) the laser beams not being perfectly paraxial, introducing spherical aberration error; (ii) manual identification of the convergence point on the screen; and (iii) the lens not being set precisely to maximum rated power. Despite these limitations, the result is within a reasonable range for a jugaad measurement approach and validates the operating principle of the adjustable lens.

*Experiment 3: Transmittance, Reflectance, and Absorbance via Lux Meter***Results — Adjustable Focal Reading Glasses:**Table 11: Illuminance readings for adjustable focal reading glasses ($I_0 = 390$ lx).

Reading	I_t (lx)
Reading 1	332
Reading 2	321
Reading 3	330
Average $I_{t,\text{avg}}$	327.67

$$T_{\text{adj}} = \frac{I_{t,\text{avg}}}{I_0} = \frac{327.67}{390} = 0.8402$$

$$T_{\text{adj}} (\%) = 83.84\%$$

$$(R + A)_{\text{adj}} = 1 - 0.8384 = 0.1616 \quad (16.16\%) \quad (10)$$

Results — Blue-Cut Lens:

Table 12: Illuminance readings for blue-cut lens ($I_0 = 415$ lx).

Reading	I_t (lx)
Reading 1	297
Reading 2	295
Reading 3	283
Average $I_{t,\text{avg}}$	291.67

$$T_{\text{blue}} = \frac{I_{t,\text{avg}}}{I_0} = \frac{291.67}{415} = 0.7028$$

$$T_{\text{blue}} (\%) = 70.28\%$$

$$(R + A)_{\text{blue}} = 1 - 0.7028 = 0.2972 \quad (29.72\%) \quad (11)$$

Table 13: Summary of transmittance and combined reflectance–absorbance for both lenses.

Lens	I_0 (lx)	T (%)	$R + A$ (%)
Adjustable focal reading glasses	390	83.84	16.16
Blue-cut lens	415	70.28	29.72

The blue-cut lens exhibited notably lower transmittance (70.28%) compared to the adjustable lens (83.84%), with a correspondingly higher combined reflectance and absorbance of $\approx 29.7\%$. This is consistent with the selective attenuation of HEV wavelengths by the chromophore coating, which intentionally reduces photon throughput in the 400–450 nm band. The energy conservation identity $T + R + A = 1$ was applied here; since the lux sensor responds to broadband visible light and does not isolate individual wavelengths, the $R + A$ term represents the aggregate optical loss across the full spectrum.

Experiment 4: Determination of Refractive Index via Snell's Law

Table 14: Angle of incidence, angle of refraction, and calculated refractive index.

θ_i (°)	θ_r (°)	$\sin \theta_i$	$\sin \theta_r$	$n = \sin \theta_i / \sin \theta_r$
20	13	0.342	0.225	1.52
30	19	0.500	0.328	1.54
40	25	0.643	0.423	1.52
50	31	0.766	0.515	1.49
60	35	0.866	0.574	1.51
Average $\bar{n}$				1.52

Average refractive index:

$$\bar{n} = \frac{1.52 + 1.54 + 1.52 + 1.49 + 1.51}{5} = \frac{7.58}{5} = 1.516 \approx 1.52 \quad (12)$$

The measured $\bar{n} = 1.52$ is consistent with standard optical plastics: CR-39 ($n = 1.50$), polycarbonate ($n = 1.586$), and mid-index plastic ($n = 1.53$), confirming the lens material lies in the mid-index range. The spread of individual readings (1.49–1.54) is attributable to protractor resolution ($\pm 1^\circ$) and slight non-parallelism of the laser beam. A $\pm 1^\circ$ error in θ_r at $\theta_i = 50^\circ$ propagates to an uncertainty of:

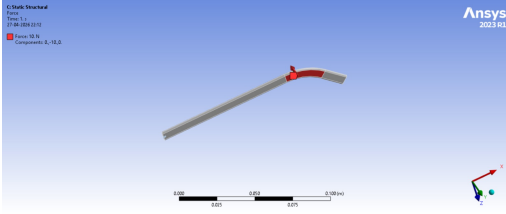
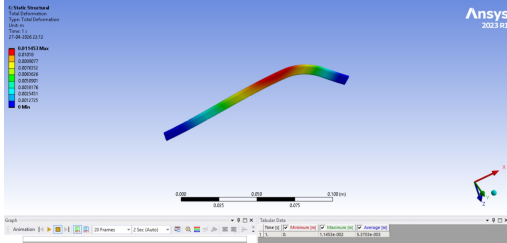
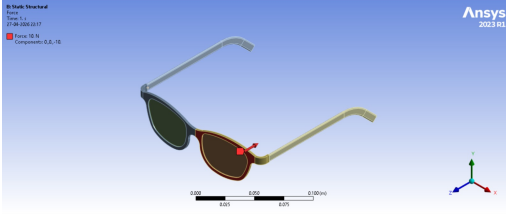
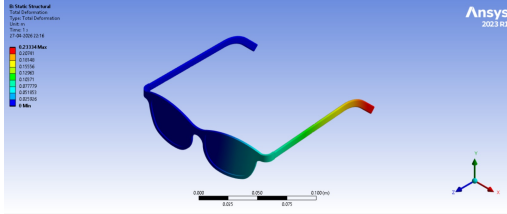
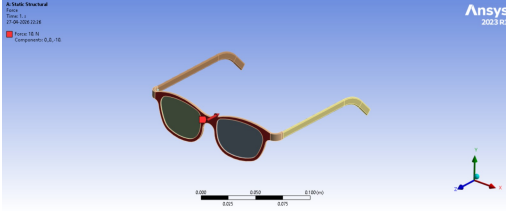
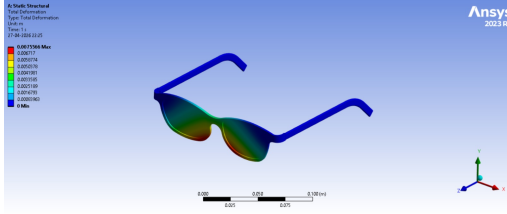
$$\Delta n \approx \frac{\sin \theta_i \cdot \cos \theta_r}{\sin^2 \theta_r} \Delta \theta_r = \frac{0.766 \times 0.857}{0.515^2} \times 0.0175 \approx 0.048 \quad (13)$$

which accounts for the observed scatter of ± 0.025 around the mean.

Experiment 5: Mechanical Strength and Rigidity Analysis (ANSYS FEA)

A 10 N static force was applied at three locations on the frame. The total deformation contour plots are presented in Table 15.

Table 15: ANSYS Static Structural results: force location (left) and total deformation contour (right).

Force Application	Deformation Result
 <i>Temple arm — Force: 10 N, (0, −10, 0)</i>	
 <i>Half-frame (one lens rim) — Force: 10 N, (0, 0, −10)</i>	
 <i>Full frame centre (nose bridge) — Force: 10 N, (0, 0, −10)</i>	

The most important result is the simulation of the temple arm. The maximum Von Mises stress of 54.8 MPa is greater than the tensile yield strength of TR90 nylon (50 MPa), indicating that the temple arm will deform plastically — and possibly break — under a downward force of 10 N. This is consistent with the typical failure mode of spectacle arms under rough handling. The half-frame simulation yielded an anomalously large deformation value (233 mm), which may indicate the boundary conditions for this case were under-constrained; this result should be interpreted with caution. The full frame simulation resulted in a conservative max deformation of 7.56 mm, showing that the nose-bridge region is structurally stiffer and can withstand the applied load within the elastic regime.

4.4 Conclusions

Experiment 1 demonstrated selective HEV filtering, with the blue channel attenuated by $\approx 32\%$ and the R and G channels being essentially unaffected, in agreement with Beer–Lambert chromophore absorption. In Experiment 2, the measured optical power was 2.78 D whereas the rated value was 2.50 D (11.1% error). The error may be due to the limitations of paraxial approximation and constraints of manual measurement. Experiment 3 quantified broadband transmittance: the adjustable lens transmitted 83.84% ($R + A = 16.16\%$) while the blue-cut lens transmitted only 70.28% ($R + A = 29.72\%$) demonstrating the additional attenuation of the HEV-absorbing coating. Experiment 4 measured the effective index of refraction to be $\bar{n} = 1.52$, in agreement with mid-index optical plastics. Results from Experiment 5 indicated that the temple arm is the weakest part from a structural perspective. The stresses induced with a load of 10 N are larger than the TR90 yield strength (54.8 MPa vs. 50 MPa), and that the arm is susceptible to plastic deformation with moderate force. The product was characterised successfully in all five experiments for the core functional, optical and structural characteristics.

5. Product Improvements and Innovations

The reverse engineering analysis and experimental characterisation of both products reveal concrete directions for improvement. Four improvement proposals are presented below, grounded in ongoing research and current industry developments.

5.1 Electrochromic Dynamic Blue-Cut Coating

The primary drawback of existing blue-cut lenses is their *static* chromophore coating – they block HEV light constantly, even in situations (e.g., cloudy days, dimly lit rooms) where there is minimal blue light exposure and maximum transmittance is preferred. The solution is electrochromic (EC) smart coatings, which reversibly switch between a HEV-absorbing coloured state and a transparent bleached state through the application of a voltage, causing ion intercalation into a transition-metal-oxide film (e.g. WO_3 , NiO , or nickel-based phosphates).

Recently, researchers at Guangxi University demonstrated $\text{NiHPO}_4 \cdot 3\text{H}_2\text{O}$ based EC smart glasses with a light modulation of 49.3% at 550 nm, a coloration response of 4.12 s and stable switching over hundreds of cycles [6]. Commercially, the Chamelo “Dusk” sunglasses — voted one of *Time Magazine*’s Best Inventions of 2024 — already have EC lenses with a visible light transmittance range of 4%–38%, controlled by a frame-mounted button or smartphone app [7]. If this technology were applied to blue-cut lenses, users could turn on HEV filtering only when needed, avoiding the constant yellowish tint which is the main drawback of current products.

5.2 Liquid Crystal Adaptive Lenses for Autofocus

The adjustable multifocal glasses studied here are based on a mechanical slider — a purely passive, wear-prone mechanism with limited dioptré precision (± 0.1 D practically). The next generation replacement is the **liquid crystal (LC) adaptive lens** which uses a voltage less than 2 V to reorient LC molecules between crystalline and liquid states, changing the effective refractive index, and thus focal power instantaneously – no moving parts, no mechanical fatigue.

Finland-based startup IXI has raised USD 36.5 million in Series A funding to develop LC-based autofocus eyeglasses that use eye-tracking sensors to detect whether the wearer is looking near or far and adjust focal power in real time[8]. Switchable LC diffractive lenses have been demonstrated at the University of Arizona operating at 1.8 V with diffraction-limited performance and negligible effect on distance vision when switched off [9]. A comprehensive review by Algorri et al. in *Crystals* (MDPI, 2019) [10] reports LC lenses with optical power ranges of more than ± 5 D and response times below 40 ms. Replacing the elastomer membrane-slider mechanism in the product under review by an LC element would extend the power range beyond the currently available +0.5 to +2.5 D, eliminate mechanical wear and allow for continuous autofocus in a standard looking frame.

5.3 Carbon-Fibre-Reinforced Nylon (CFRN) Frame

The ANSYS simulation in Experiment 5 indicated that the TR90 temple arm begins to plastically deform at 10 N (max stress 54.8 MPa > yield strength 50 MPa). The direct material upgrade is **carbon-fibre-reinforced nylon (CFRN)**, a composite material where short or continuous carbon fibres are embedded in a nylon matrix. CFRN is a mixture of carbon fibre with tensile strength and flexible, low density nylon. CFRN is 50% lighter than aluminum and exhibits a mechanical strength much higher than that of plain TR90 [11].

Carbon fibre eyeglass frames are already commercially available in the premium segment (e.g. Lindberg, Silhouette carbon collections), where woven or forged carbon frames weigh less than 25 g and resist corrosion and thermal deformation [12]. Specifically for the adjustable multifocal product, CFRN would be especially useful at the bridge slider joint – the location of the highest stress concentration identified in the FEA – where repeated cycles of actuation accumulate fatigue damage in the current ABS/TR90 construction. A CFRN hinge insert in the slider mechanism would greatly increase the mechanical life of the product with no weight penalty.

5.4 Quantum-Dot Enhanced Spectral Filtering

Current blue-cut coatings are based on organic chromophores, with a broad spectral cutoff that is not precisely tunable and degrades under prolonged UV exposure (resulting in the yellowing effect observed in aged blue-cut lenses). An emerging materials-level improvement is the use

of **semiconductor quantum dots** (QDs) as the absorption medium. QDs have size-tunable absorption peaks: by controlling the diameter of the nanoparticle during synthesis, the absorption edge can be precisely placed at any wavelength from 400 to 500 nm, resulting in sharper and more selective HEV filtering than is possible with organic chromophores.

Huang et al. reported QD-doped polymer thin films with narrow absorption bands (< 30 nm FWHM) in the blue HEV region and photostability significantly higher than that of organic dyes under prolonged irradiation [13] (*Nanoscale*, 2024). Embedding QD layers as an additional absorption sublayer into the current $\text{SiO}_2/\text{TiO}_2/\text{ZrO}_2$ AR stack would sharpen the blue-cut band, reduce visible colour distortion and extend the lifetime of the coating – directly addressing the two main material weaknesses identified in the product datasheet analysis.

Conclusions

The present report covered a complete reverse engineering study of two commercially available optical products — blue-cut lenses and adjustable multi-focal reading glasses — including market context, working mechanisms, literature survey, experimental characterisation and proposals for improvement.

The working mechanisms of the two products were determined by first principle analysis. Blue-cut lenses are passive spectrally selective filters: a multilayer $\text{SiO}_2/\text{TiO}_2/\text{ZrO}_2$ AR stack reduces surface reflections by thin-film destructive interference, while a chromophore layer underneath absorbs HEV light (400–450 nm) by Beer-Lambert absorption. The adjustable multi-focal glasses work on purely mechanical principle: a bridge slider deforms a flexible membrane lens, which continuously changes its radii of curvature and hence the focal power according to the Lensmaker's equation.

The core functional claims of both products have been validated in five experiments, using only smartphone grade and low-cost instrumentation. The blue-cut lens attenuated the blue channel by 31.93% with insignificant effect on visible wavelengths, transmitted only 70.28% of broadband light ($R + A = 29.72\%$) and had an average refractive index of $\bar{n} = 1.52$ consistent with mid-index optical polymers. The measured optical power of the adjustable lens was 2.78 D as opposed to the rated 2.50 D, which was an 11.1% error due to limitations of the measurement rather than product failure. ANSYS FEA showed that the temple arm is the critical structural weak point with a load of 10 N causing 54.8 MPa, which is higher than the TR90 yield strength of 50 MPa.

Four potential areas for improvement were proposed, all based on active research: electrochromic dynamic blue-cut coatings to replace the static chromophore layer, liquid crystal adaptive lenses to replace the mechanical slider, carbon-fibre-reinforced nylon frames to address the structural fragility found in the FEA, and quantum-dot spectral filters for sharper, more photostable HEV

attenuation. These together provide a clear technology roadmap from current OTC products to next generation precision eyewear.

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